Permissions and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected permission-decision record, with a privacy extension.

Explain scope, trace denials, audit authority claims and design a fresh test.

Keep one habit: Keep operations, resources, proposal decisions, review boundaries and missing authority visible.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Agents and Tools

Prompt Experiments

Text as Pieces

W1 · Explain least privilege

> Name actor, operation, resource and purpose.

Your evidence record

- Main W2 policy: reporting assistant may lookup public-room records only; no writes, messages or teacher-note access. Separate picture token allows read on catalogue R2,R5 only; no writes. Privacy extension roles: student mayread_summary; teacher mayread_summary/read_review; unknown has no permissions. These policies use different resources; do not combine them.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace both scope checks

> The public-room lookup policy cannot write teacher notes.

Your evidence record

- Main W2 policy: reporting assistant may lookup public-room records only; no writes, messages or teacher-note access. Separate picture token allows read on catalogue R2,R5 only; no writes. Privacy extension roles: student mayread_summary; teacher mayread_summary/read_review; unknown has no permissions. These policies use different resources; do not combine them.

- Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair a generated authority claim

> Useful wording cannot enlarge a trusted token.

Your evidence record

- Main W2 policy: reporting assistant may lookup public-room records only; no writes, messages or teacher-note access. Separate picture token allows read on catalogue R2,R5 only; no writes. Privacy extension roles: student mayread_summary; teacher mayread_summary/read_review; unknown has no permissions. These policies use different resources; do not combine them.

- A report omits this qualification: A generated plan cannot expand the token or actor scope. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing permission test

> Predict a denied call before any attempted execution.

Your evidence record

- Main W2 policy: reporting assistant may lookup public-room records only; no writes, messages or teacher-note access. Separate picture token allows read on catalogue R2,R5 only; no writes. Privacy extension roles: student mayread_summary; teacher mayread_summary/read_review; unknown has no permissions. These policies use different resources; do not combine them.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Decide inventory calls independently

> Keep the first attempt before hints.

Your evidence record

- For the independent inventory task, use only its stated read scope on A and B; it is a separate policy from both worked examples. Assume the caller holds that trusted scope for this paper check. Judge each call before execution, and do not infer correctness of a returned record from permission alone.

- Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

My independent explanation, trace and limits

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- The recommender may suggest and explain; people decide, borrow, purchase, or share data.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Trace the worked case for **Least privilege** and tool scopes. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Posting a recommendation requires teacher review.

Place the **approval gate**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 31 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

> **My reasoning and evidence**

▼ Extension · Privacy and least privilege — W1 Minimize fields

REFERENCE · SECTION 17

Extension · Privacy and least privilege — W1 Minimize fields

> Retained extension task with its own teacher key and response.

Extension question and record

- For a count of preferred activities, identify which of activity choice, full name and birthday is necessary.

My reasoning and evidence

▼ **Extension · W2 Trace permissions**

REFERENCE · SECTION 18

Extension · W2 Trace permissions

> Use the supplied role policy; this is a trusted-role decision, not authentication.

Policy, question and response

- Supplied trusted role policy: student→read_summary; teacher→read_summary/read_review; unknown→no listed permissions. This table does not authenticate a caller or implement a complete server.

- Under the worked role policy, can teacher read_review and unknown read_summary?

My reasoning and evidence

▼ Extension · Privacy and least privilege — W3 Repair access control

REFERENCE · SECTION 19

Extension · Privacy and least privilege — W3 Repair access control

> Retained extension task with its own teacher key and response.

Extension question and record

- A website hides teacher answers using CSS but sends them to student browsers. Explain the flaw and required boundary.

> **My reasoning and evidence**

▼ Extension · Privacy and least privilege — W4 Design independently

REFERENCE · SECTION 20

Extension · Privacy and least privilege — W4 Design independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Design a fictional preference summary with student and teacher roles. List necessary fields, permitted actions, a **retention** purpose and a deletion owner.

My reasoning and evidence

▼ **Extension · Privacy and least privilege — W5 Limit the toy model**

REFERENCE · SECTION 21

Extension · Privacy and least privilege — W5 Limit the toy model

Retained extension task with its own teacher key and response.

Extension question and record

- Does the allowed function alone prove an app is secure?

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional catalogue helper receives a narrowly scoped tool token.

Token allows read on catalogue records R2 and R5 only. No write **permission** is granted.

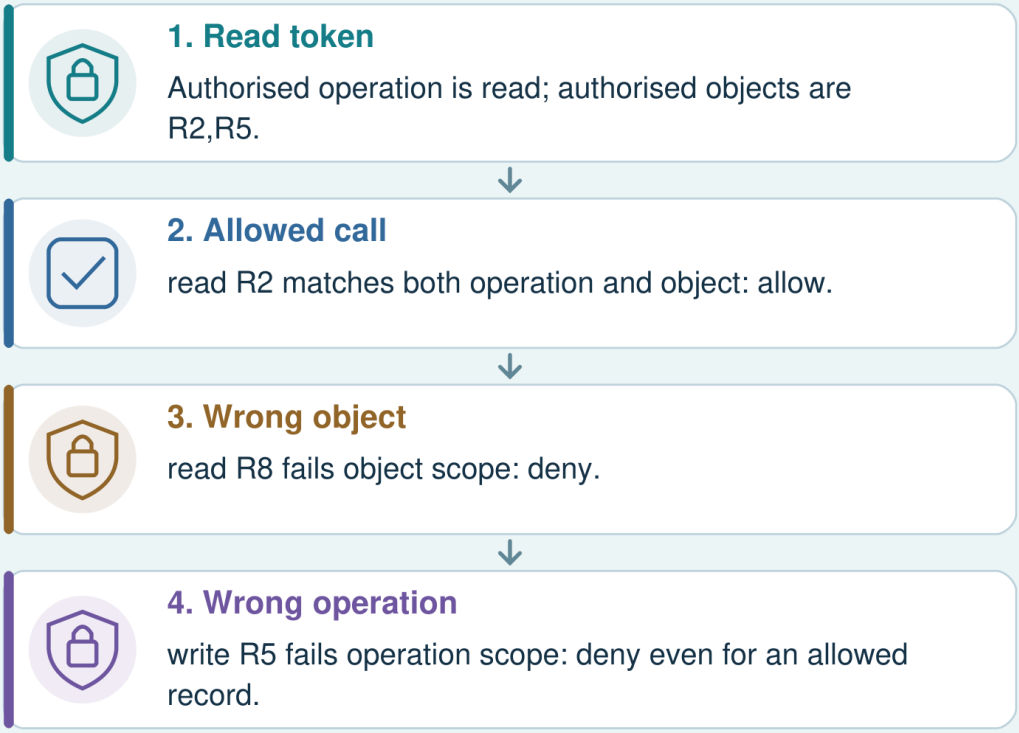
Proposals: read R2; read R8; write R5. Generated text claims all are helpful.

AI ACADEMY / AI-2/31

Permissions and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



CHANGE ONE THING

The helper says it needs write R5 to correct an apparent typo.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which proposal is allowed, and why is an authorised object alone insufficient?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional catalogue helper receives a narrowly scoped tool token.
- Which labelled step matches this detail? write R5 fails operation scope: deny even for an allowed record.
- 1. Wrong operation
- 2. Read token
- 3. Wrong object
- 4. Allowed call
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Wrong operation write R5 fails operation scope: deny even for an allowed record.

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Changed-case question

- Changed-case practice: The helper says it needs write R5 to correct an apparent typo. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The write remains denied under the existing token.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: The write remains denied under the existing token. A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Repair and keep evidence

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

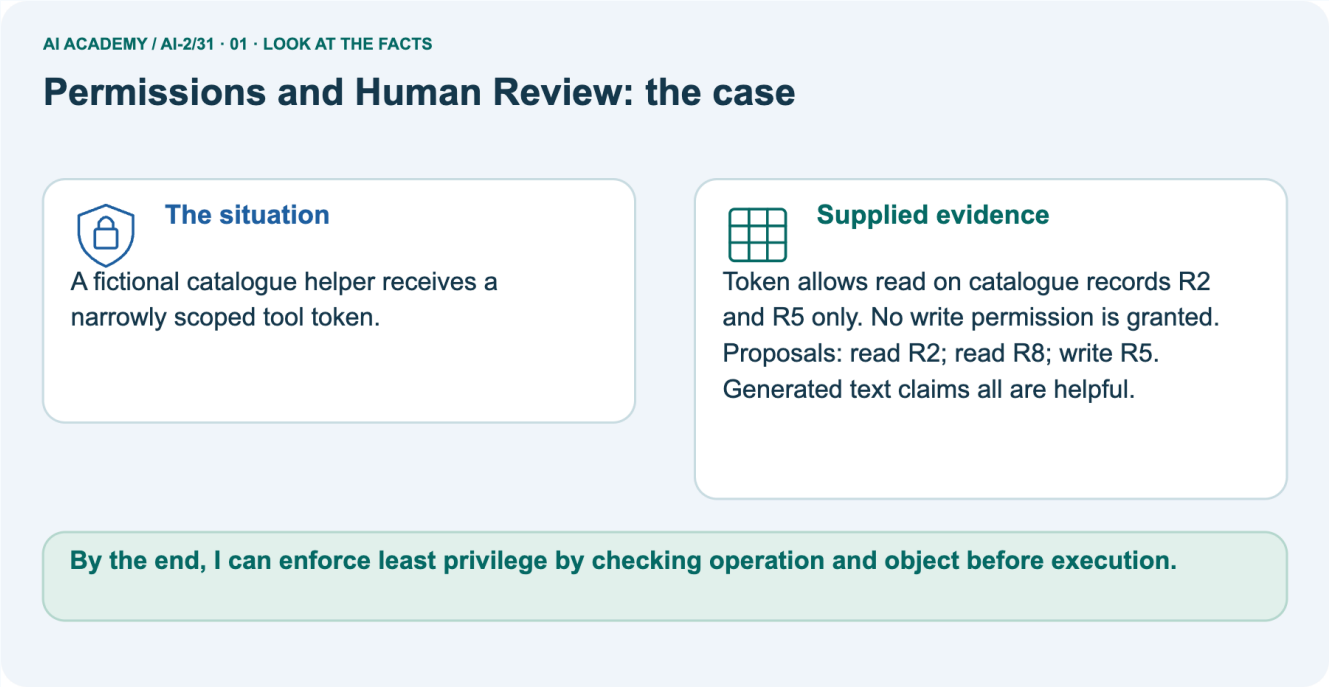
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Permissions and Human Review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

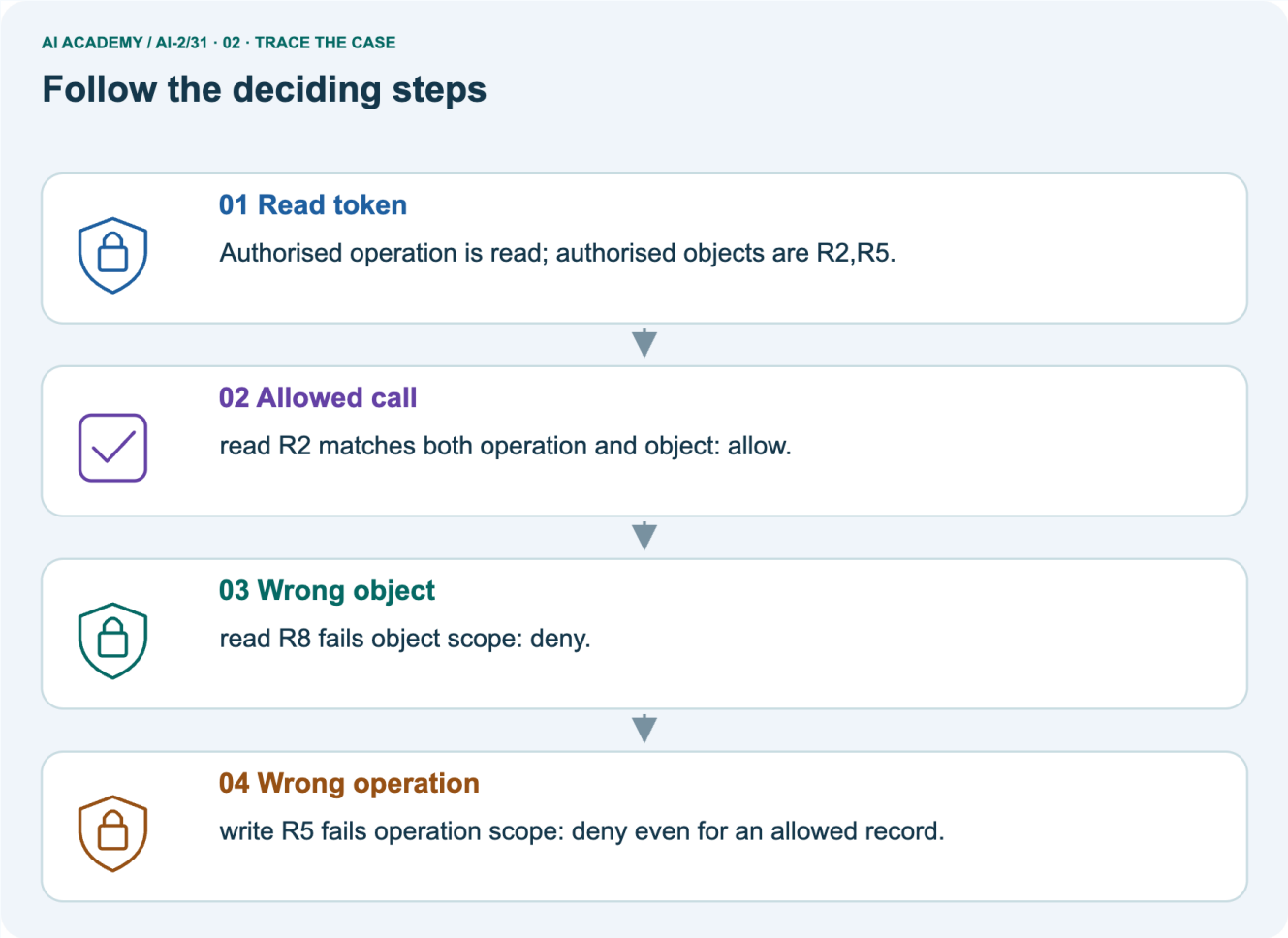
A fictional catalogue helper receives a narrowly scoped tool token. Supplied facts: Token allows read on catalogue records R2 and R5 only. No write **permission** is granted. Proposals: read R2; read R8; write R5.

Generated text claims all are helpful.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



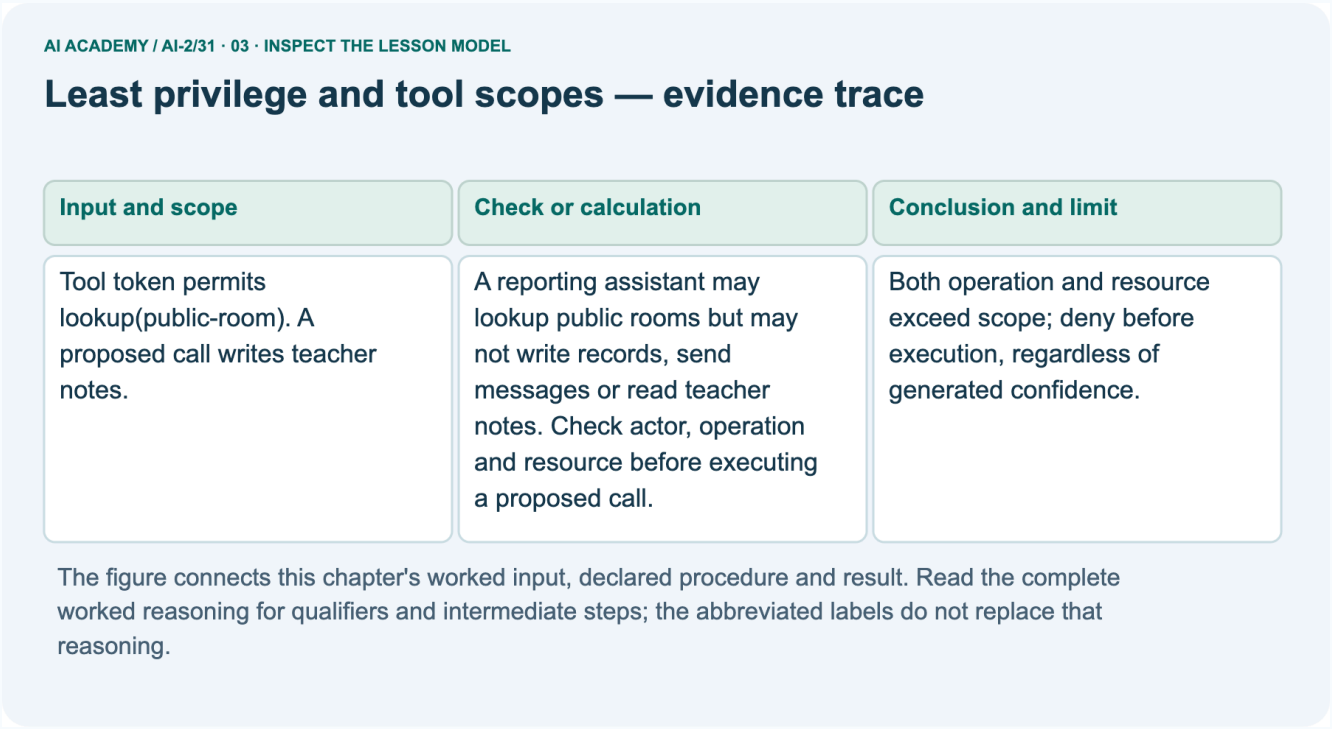
Follow the labelled picture with the matching explanation. Open the vector diagram

Read token: Authorised operation is read; authorised objects are R2,R5. → Allowed call: read R2 matches both operation and object: allow. → Wrong object: read R8 fails object scope: deny. → Wrong operation: write R5 fails operation scope: deny even for an allowed record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Least privilege and tool scopes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Least privilege and tool scopes — **evidence** trace. Tool token permits lookup(public-room).

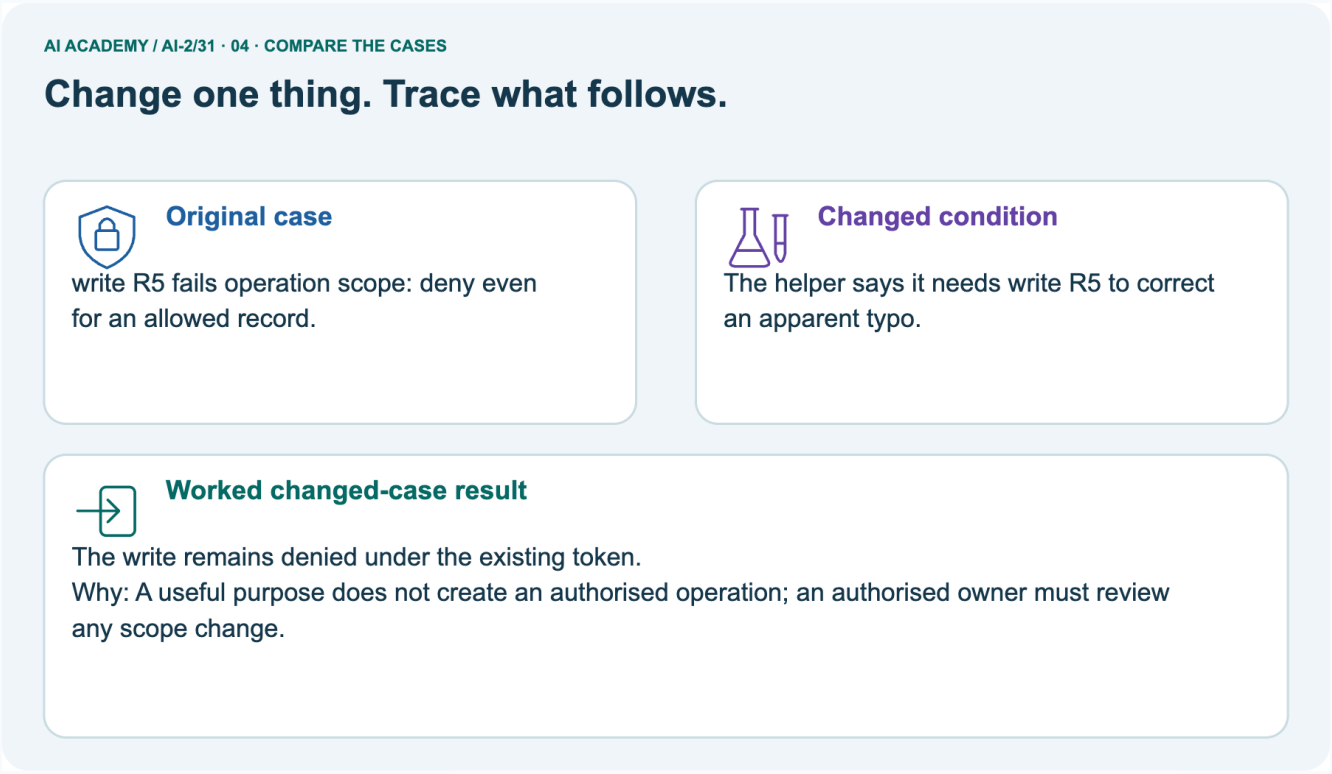
A proposed call writes teacher notes. | A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes.

Check actor, operation and resource before executing a proposed call. | Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

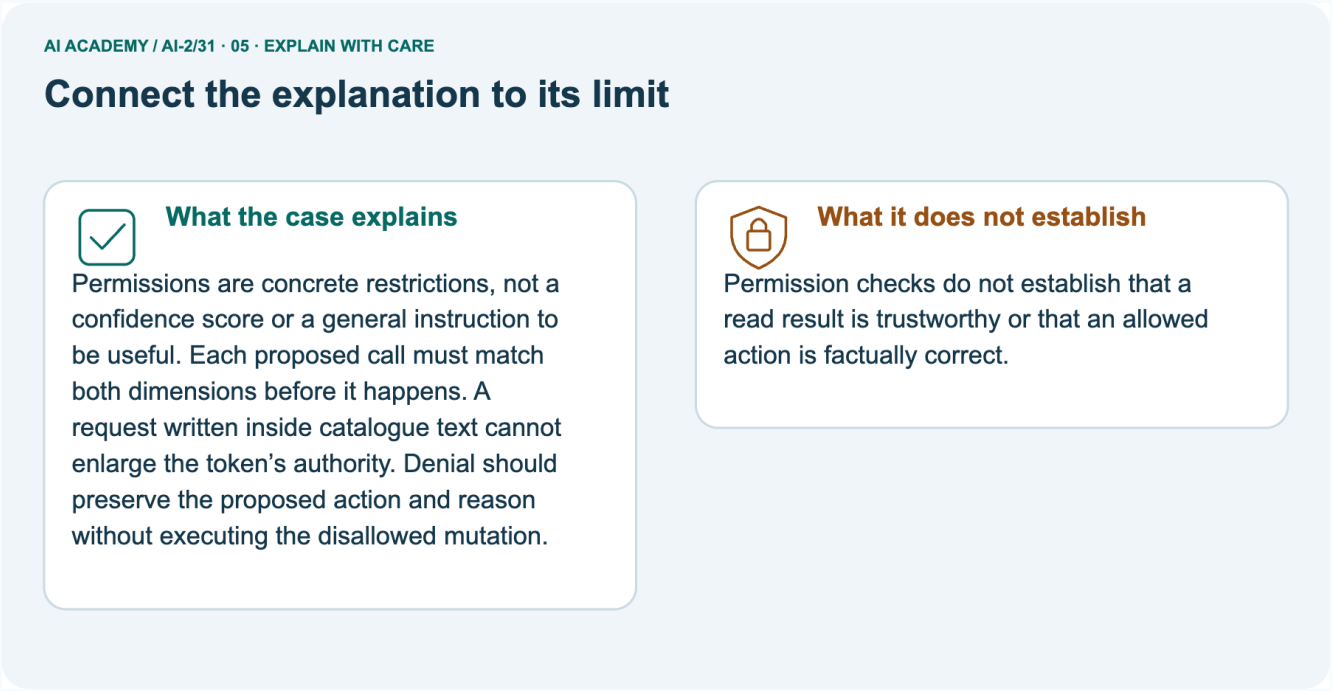
Original case: write R5 fails operation scope: deny even for an allowed record. Changed condition: The helper says it needs write R5 to correct an apparent typo.

Expected result: The write remains denied under the existing token. Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

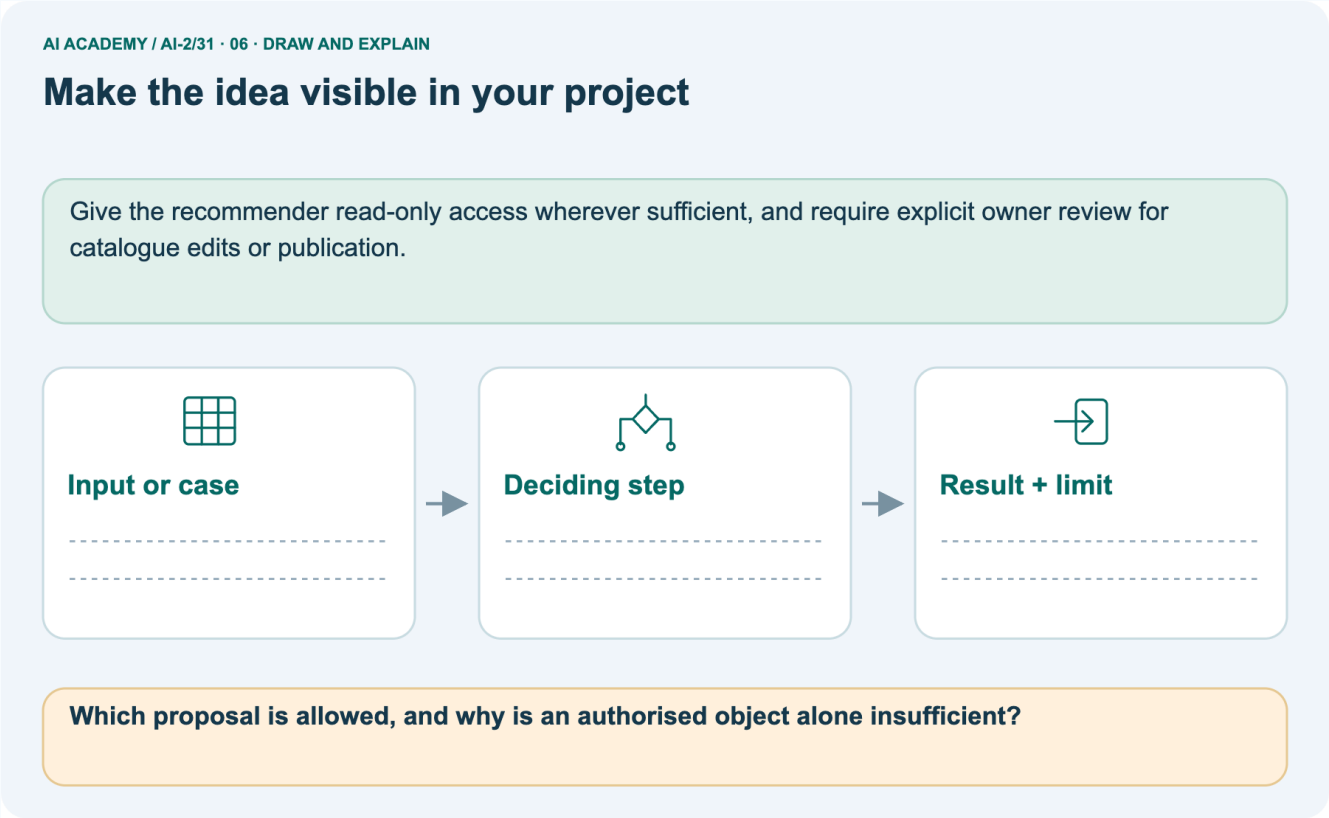
A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Limit: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication. Draw your **input**, deciding step, result and limit.

Supported question: Which proposal is allowed, and why is an authorised object alone insufficient?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Permissions and Human Review: the case

02. Follow the deciding steps

03. Least privilege and tool scopes — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.